

APIs Deep Dive: Query, Admin, Ingestion, Analytics

Public Query API: GET /v1/query

Purpose: Retrieve autocomplete suggestions for user queries.

Method: GET /v1/query?tenantId=...&indexId=...&q=...&limit=10&locale=en

Parameters:

- tenantId: Tenant identifier
- indexId: Target index
- q: Input query string
- limit: Max results
- locale: Language/region

Response:

```
{
  "results":[{"id":"e1","title":"example","score":0.91}],
  "timings":{"prefixMs":7,"rankMs":8,"totalMs":20}
}
```

Notes: Supports explain=true for debugging recall and ranking breakdowns.

Latency target: <80ms backend.

Admin APIs: Manage Indices, Synonyms, Rules

Purpose: Lifecycle management of indices and search behaviors.

Endpoints:

- POST /v1/indices (create index)
- PATCH /v1/indices/{id} (update index settings)
- POST /v1/indices/{id}/synonyms (add synonym set)
- POST /v1/indices/{id}/rules (add ranking/business rules)
- DELETE /v1/indices/{id} (delete index)

Usage: Allow operators to configure schemas, synonyms, and rules to tune search relevance.

Security: Requires admin API key with audit logging.

Ingestion APIs: Batch Upserts, Deletes, Streaming

Purpose: Load and update data into the autocomplete index.

Endpoints:

- POST /v1/entities:batchUpsert (bulk create/update entities)
- DELETE /v1/entities/{id} (delete entity)
- POST /v1/ingestion/stream (streaming ingestion of entity updates)

Payload example:

```
{ "entities": [{"id": "e1", "title": "Sample", "popularity": 0.8}] }
```

Guarantees: At-least-once ingestion with idempotent keys; stream cursors support checkpointing.

Use cases: Keep indices fresh with product catalog updates, logs, or event streams.

Analytics APIs: Query Events, Selection Events, Reports

Purpose: Track usage, measure KPIs, and generate reports.

Endpoints:

- POST /v1/events/query (log query event)
- POST /v1/events/selection (log click/selection)
- GET /v1/analytics/reports?indexId=... (retrieve CTR, abandonment, latency distributions)

Event payload example:

```
{ "query": "lpto", "resultCount": 10, "latencyMs": 22, "timestamp": "..." }
```

Reports: Provide aggregated analytics for tenants, driving ranking model improvements.

Notes: Analytics events sampled and PII-truncated for privacy compliance.